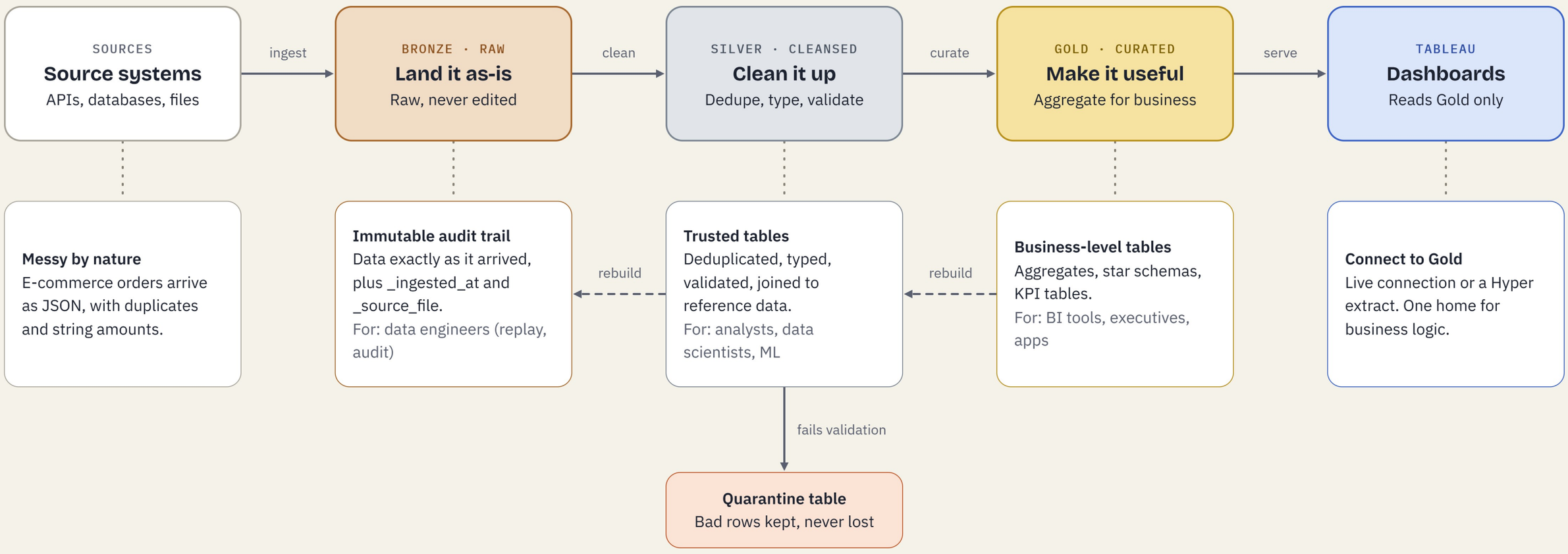
Medallion Architecture + Tableau APIs

From raw data to trusted dashboards: three layers in the lake, seven APIs at the last mile.

01 Medallion architecture

Every hop adds quality. Bronze is never edited, so any layer can be rebuilt from the one before it.



02 The Tableau API family

Not one API but a family, and each answers a different question. Names and versions change, so check Tableau's developer docs before building.

ADMINISTER
REST API
Administer Tableau Server and Cloud.
Publish workbooks, manage users, trigger extract refreshes.

LINEAGE
Metadata API
Query metadata and lineage with GraphQL.
Which dashboards break if I change this column?

EMBED
Embedding API v3
Embed views in your own web apps.
A <tableau-viz> web component in your portal.

EXTEND
Extensions API
Build custom dashboard objects.
Custom widgets that live inside a dashboard.

EXTRACTS
Hyper API
Create and read .hyper extracts in code.
Push Gold data into an extract without Desktop.

CONNECT
Connector SDK
Build custom data connectors.
Reach a source Tableau doesn't support.

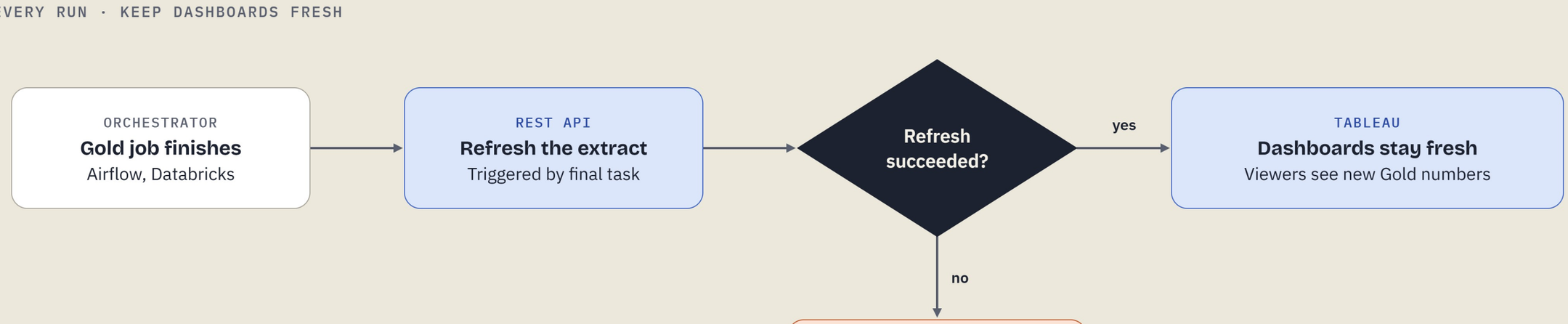
EVENTS
Webhooks
Event notifications.
Alert the team when an extract refresh fails.

AUTH TIP
Personal Access Tokens
Revocable and built for automation, unlike passwords.
In Python, use tableauserverclient (TSC).

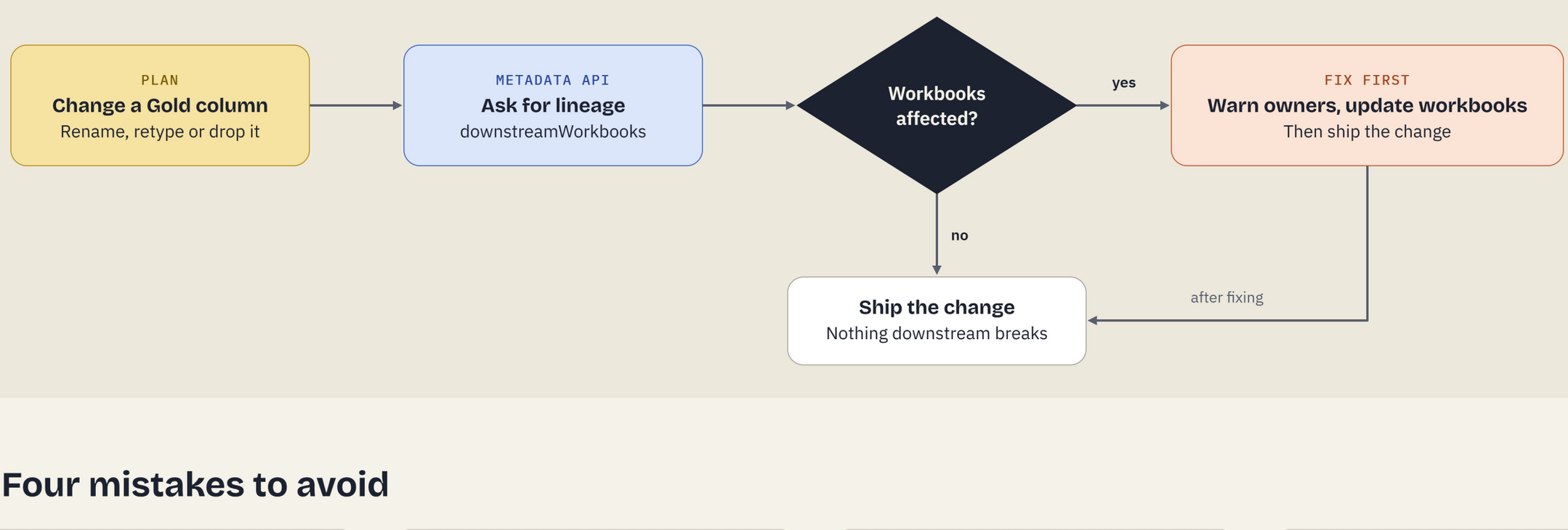
03 How the two fit together

Orchestration keeps dashboards fresh, and lineage tells you what breaks before you change a Gold column.

EVERY RUN · KEEP DASHBOARDS FRESH



BEFORE A SCHEMA CHANGE · KNOW WHAT BREAKS



04 Four mistakes to avoid

MISTAKE 1
Fixing data in Bronze
It destroys your audit trail. Clean in Silver instead.

MISTAKE 2
Pointing Tableau at Silver
Slow queries, and every dashboard reinvents the business logic. Read Gold.

MISTAKE 3
Exactly three layers, always
Real systems add sub-layers, and pure ML work may skip Gold.

MISTAKE 4
No quality gates
Rows that fail Silver checks go to quarantine. They never just vanish.